

Urban Expansion Assessment Document

Al-Seh within Al-Ahsa Governorate

Document purpose: This document presents a planning assessment of urban expansion requirements for **Al-Seh** within **Al-Ahsa**, with emphasis on population growth, land-supply adequacy, spatial form, planning context, and the most supportable direction for controlled future expansion [1] [2] [3].

General Overview

Al-Seh is experiencing a clear phase of urban growth within the wider Al-Ahsa context. Current indicators show that population growth has accelerated more rapidly than the increase in developable urban land. This has led to a marked rise in density and a stronger degree of pressure on the existing urban footprint. On this basis, the settlement should be treated as a local planning case requiring controlled and structured spatial response rather than simple continuation of the current development pattern [1] [2].

The available planning reading indicates that Al-Seh remains suitable for a formal urban-expansion assessment because the evidence combines demographic change, land-supply conditions, visible spatial fragmentation, and a recognizable planning envelope. The main planning issue is therefore whether the current developable area is sufficient to accommodate projected demand efficiently, or whether an organized boundary adjustment is justified [1] [2] [3].

Location and Planning Context

Item	Assessment statement
Settlement	Al-Seh
Administrative context	Located within Al-Ahsa
Assessment focus	Urban growth pressure, land-supply adequacy, and controlled expansion need
Planning horizon reflected in available material	Current comparative reading based on 2020 and 2025 indicators, with planning implications extending into the next development stage [1]

Primary planning question	Whether the existing developable footprint remains sufficient, or whether structured expansion should be adopted [1] [2]
Current planning conclusion	Controlled expansion is justified, with westward growth remaining the strongest current option [1] [2] [3]

Demographic and Regional Studies

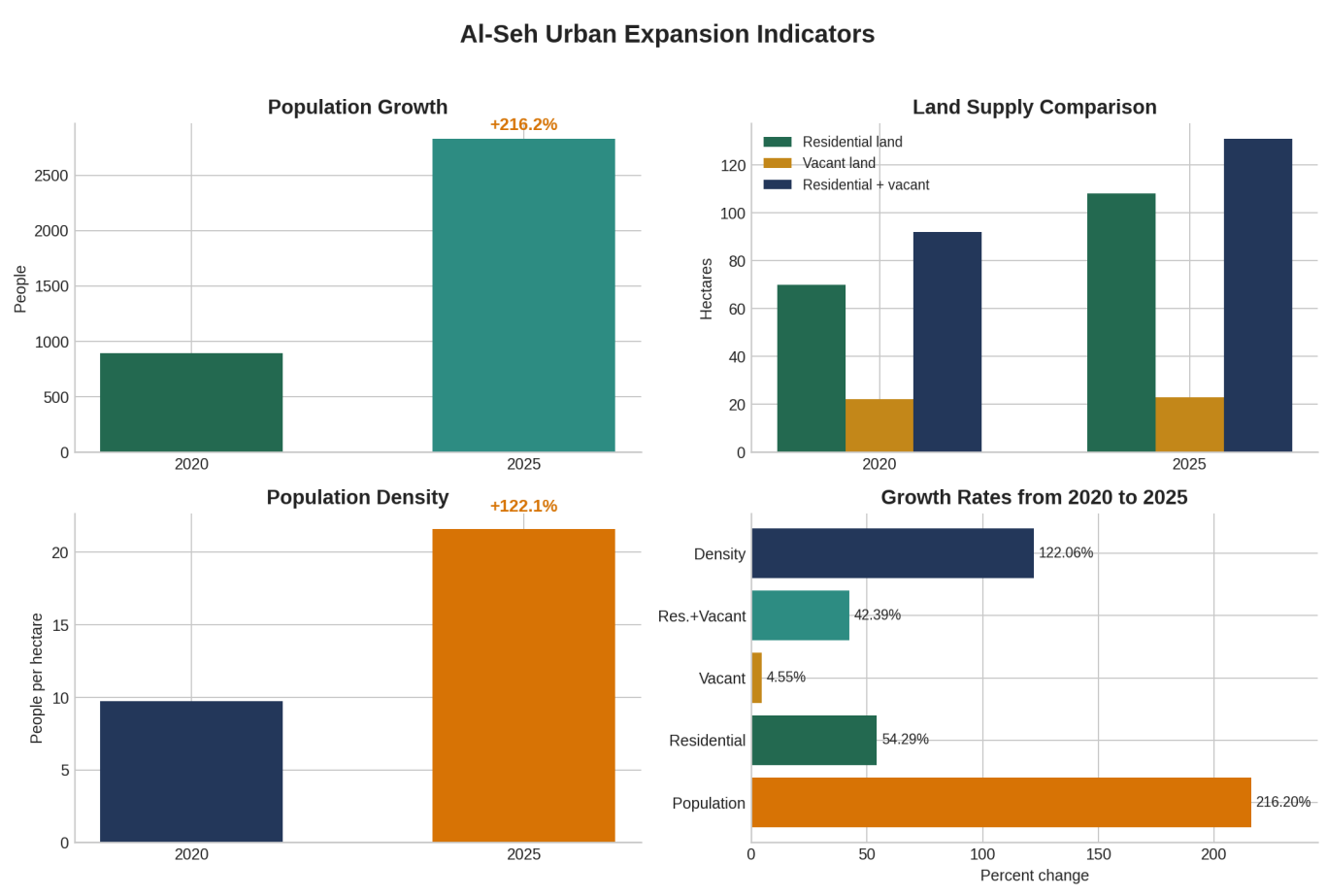
The demographic indicators show that Al-Seh is absorbing growth at a pace that exceeds the rate of land-supply expansion. Population increased from **895** in 2020 to **2,830** in 2025, equivalent to **216.20% growth** over the comparison period [1]. This level of change is material from a planning perspective and cannot be treated as marginal growth.

At the same time, the urban land indicators show a much slower increase. Residential land rose from **70 hectares** to **108 hectares**, while the combined residential-plus-vacant area increased from **92 hectares** to **131 hectares**. Vacant land changed only from **22 hectares** to **23 hectares**, indicating that reserve capacity expanded only slightly despite strong population growth [1].

Population and land indicator	2020	2025	Change	Planning implication
Population	895	2,830	+216.20%	Strong increase in demand [1]
Residential land	70 ha	108 ha	+54.29%	Supply grew, but far below population growth [1]
Vacant land	22 ha	23 ha	+4.55%	Very limited increase in reserve land [1]
Residential + vacant land	92 ha	131 ha	+42.39%	Developable footprint expanded less than demand [1]

Density	9.73 p/ha	21.60 p/ha	+122.06%	Urban pressure intensified sharply [1]
---------	-----------	------------	----------	--

The derived ratios strengthen this reading. Population increased by approximately **3.16 times**, whereas the residential-plus-vacant footprint increased by only **1.42 times**. Density rose by **11.87 persons per hectare**, which indicates that a substantial share of growth has been absorbed through intensification inside the existing urban footprint rather than through balanced outward expansion [1].



Urban Studies

The urban studies reading indicates that the settlement is operating under real growth pressure. The key issue is not merely that population has risen, but that the available developable land has not expanded at a comparable pace. This creates a structural imbalance between demand and supply and supports the need for a controlled planning response [1].

Urban studies item	Value / finding	Planning reading
--------------------	-----------------	------------------

Current density	21.60 persons/ha in 2025 [1]	High increase compared with the 2020 base condition
Base density	9.73 persons/ha in 2020 [1]	Indicates that current pressure is materially higher than the earlier period
Residential land available in 2025	108 ha [1]	Expanded, but not at a rate sufficient to match population growth
Vacant land available in 2025	23 ha [1]	Limited reserve capacity for flexible future accommodation
Total residential + vacant footprint	131 ha [1]	Current developable envelope remains under pressure
Indicative additional requirement	Approximately 15 ha westward [1] [2]	Most supportable current expansion scale and direction
Assessment confidence	Moderate [1] [2]	Suitable at assessment level; final implementation requires validation

Spatial Form and Urban Pattern

The spatial evidence indicates that Al-Seh does not currently display a fully consolidated urban form. The visible built-up pattern is fragmented, with discontinuous clusters rather than uniform occupancy across the wider planning envelope. At the same time, the surrounding boundary framework indicates that the settlement is already being viewed within a broader spatial logic rather than as an isolated built patch [1] [3].

This spatial relationship is significant because it supports organized expansion rather than unstructured sprawl. The pattern suggests that growth pressure is real, yet the current form of occupation remains uneven. Accordingly, a controlled extension aligned with planning boundaries is more appropriate than continued ad hoc intensification or scattered outward spread [1] [3].

Spatial observation	Interpretation	Planning significance
----------------------------	-----------------------	------------------------------

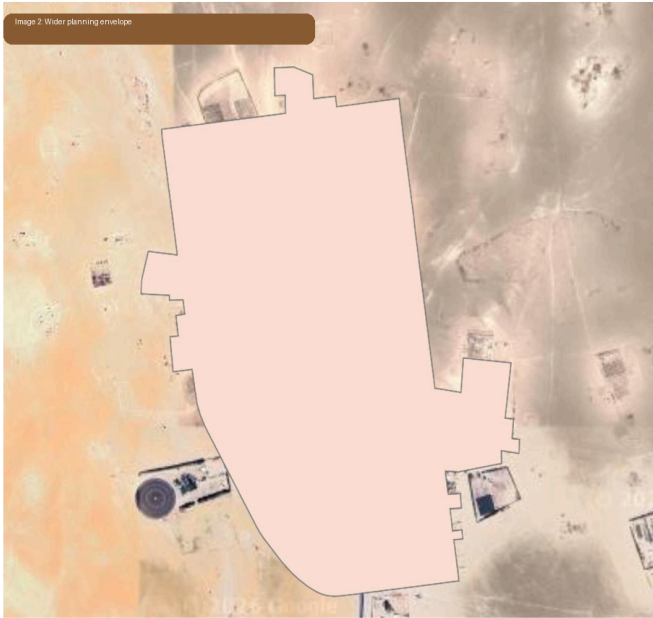
Fragmented built-up clusters	Development is unevenly distributed across the local footprint [3]	Indicates incomplete consolidation of the urban mass
Broader outer envelope	A larger planning frame exists around the existing built-up areas [3]	Supports structured urban-growth discussion
Gap between occupied form and boundary logic	Not all potential space has been occupied in a uniform way [3]	Strengthens the case for organized planning intervention

AI Seh Visual Boundary Comparison

Observed built-up pattern compared with the wider planning envelope from the uploaded reference images



Fragmented urban clusters suggest current development to be discontinuous and concentrated along a limited corridor.



A broader enclosing boundary indicates room for structured expansion within a planned development envelope.

Planning and GIS Context

The GIS review confirms that the settlement is supported by a relevant planning-data environment. The reviewed layers include parcel data, built-up urban mass, the urban boundary, the development boundary, and the official planning boundary. The parcel layer

contains **1,174 polygon records** and includes land-use information such as residential and public-service classifications [2].

The planning-boundary metadata identifies the local plan as "**تخطيط هجرة السيح**", plan number **4/646**, associated with **المunicipality الجفر**. This gives the assessment a formal planning anchor and supports treatment of the settlement within a recognized regulatory framework [2].

Layer	Key note	Planning value
الأراضي.shp	1,174 parcel polygons with land-use attributes [2]	Supports land-use interpretation and parcel-level review
الكتلة_العمرانية.shp	Built-up mass layer [2]	Represents the current urbanized footprint
حدود_المخطط.shp	Official planning-boundary metadata [2]	Links the assessment to formal planning context
الحيز_العمراني_1450.shp	Urban boundary layer [2]	Supports interpretation of the current urban extent
حد_التنمية.shp	Development boundary layer [2]	Supports future growth-envelope review

A technical note remains necessary. The reviewed GIS layers are not all stored in a single coordinate system. Some are in **WGS84 geographic coordinates**, while others are in **WGS84 / UTM Zone 39N**. For that reason, any engineering-grade overlay, area calculation, or implementation-stage mapping should first complete coordinate-system harmonization and geometry validation [2].

Preferred Direction of Expansion

Within the currently available evidence, the strongest expansion direction remains the **west**. The planning notes indicate that the **east** is more constrained by the presence of industrial uses, while the **north** and **south** are relatively weaker in terms of available public-service support. By comparison, the **west** remains the least constrained option under the current evidence base and therefore the most supportable direction for controlled growth [1].

Direction	Current reading	Planning implication
-----------	-----------------	----------------------

East	Industrial facilities present [1]	May require compatibility review and buffering
North	Public-service support is more limited [1]	May require stronger service reinforcement
South	Similar service limitation to the north [1]	Also less favorable under current conditions
West	Strongest current planning option [1] [2]	Most supportable direction for structured expansion

The currently supportable indicative scale is **approximately 15 hectares** in the westward direction. This should be read as a planning-level recommendation rather than a final implementation commitment. Final designation remains subject to service review, infrastructure adequacy, land suitability, and engineering validation [1] [2].

Regulatory and Technical Notes

The current assessment supports a planning conclusion at document level, but implementation should proceed only after completion of the standard technical and institutional reviews. These include infrastructure and service-capacity verification, access and connectivity review, land-suitability screening, and GIS preprocessing to ensure reliable overlay and area calculations [1] [2].

Required follow-up item	Purpose
Infrastructure and service review	Confirm support capacity for any future western expansion
Land-suitability screening	Identify feasible sectors and exclude unsuitable areas
Access and network review	Ensure physical connectivity and operational viability
CRS harmonization and geometry validation	Support accurate mapping and area calculations
Planning approval review	Maintain statutory and governance compliance

Summary Statement

The current planning indicators for **Al-Seh within Al-Ahsa** support a clear conclusion. Population and density have increased much faster than the available developable land supply, while the spatial evidence shows a fragmented built-up pattern inside a broader planning envelope. These conditions support the need for a **controlled urban-expansion response** rather than continued reliance on the current footprint alone [1] [3].

On the basis of the reviewed evidence, the most supportable current planning direction remains **westward expansion of approximately 15 hectares**, subject to the technical, engineering, and planning validations required before final implementation [1] [2].